

Agent-based simulation for pedestrian evacuation: A systematic literature review

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ABSTRACT

Agent-based models (ABMs) offer promise for realistically simulating human behaviours and interactions during emergency evacuations. This review aims to systematically assess the state of the art in ABM-based evacuation modelling with respect to methodologies, validation practices, and the associated challenges over the past decade. The review critically examines 134 studies from 2013 to 2023 that have applied ABMs for pedestrian evacuation simulation to synthesise current capabilities, limitations, and advancement pathways. Findings identify persistent challenges related to modeller bias, computational complexity, data scarcity for calibration and validation, and the predominance of simplistic rule-based decision-making models, while promise exists with the adoption of flexible behavioural frameworks, high-performance computing architectures, machine learning techniques for adaptive agent behaviours and surrogate modelling, and evolutionary computation methods for transparent rule generation. The findings underscore the importance of interdisciplinary collaboration among behavioural scientists, modellers, and emergency planners to enhance the realism and reliability of ABMs. By providing a critical synthesis of the state-of-the-art and proposing future research directions, this review aims to accelerate the development and application of ABMs that can meaningfully enhance the safety and resilience of communities facing emergencies.

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1. Introduction

Computer simulation using Agent-Based Models (ABMs) has been an effective means of exploring emergent behaviour patterns in complex systems [1–4]. ABMs use an object-oriented modelling technique to simulate the actions and interactions of individual autonomous decision-making entities, evaluating the effect of heterogeneity of individuals as a whole to capture emerging phenomena [5]. It has found applications in diverse fields, spanning from financial stock market behaviours [6] to medical research [7,8]. It has also become established in the social sciences to study emergency evacuation behaviour [9] where pedestrian simulation has become critical to understanding building environments, urban spaces and transport emergencies [10,11].

Pedestrian evacuation modelling using ABMs allows for a detailed representation of individual pedestrians as autonomous agents, each characterised by distinct attributes, decision-making rules, and interactions. Agents can be programmed with attributes such as walking speed, reaction time, familiarity with the environment, decision-making heuristics, as well as demographic features such as age and gender. Further, ABMs can incorporate factors like panic, herding behaviour, and bottlenecks within the environment that significantly impact evacuation behaviour. The agents interact with each other and the built environment, navigating spaces while avoiding obstacles and responding to cues like signage and crowd density [12,13]. These individual cognitive processes, social dynamics, movement physics and environmental interactions enable ABMs to capture complex collective phenomena emerging from the bottom up. Simulations using ABMs provide valuable insights into evacuation times, congestion points, and the effectiveness of different evacuation strategies or building designs [14]. However, significant challenges persist around accurately modelling human decision-making under uncertainty, balancing behavioural fidelity with computational tractability, acquiring empirical micro-level data for calibration, and establishing model credibility [15–17].

Dynamic pedestrian movement analyses can assist in effectively planning and designing cities, transport networks and people movement with building structures. In recent years, a significant amount of resources has been invested in researching evacuation behaviour simulation using experimental data combined with Artificial Intelligence (AI) related computational approaches [18–20]. However, the prediction of outcomes using human interpretable AI and understanding of the inner mechanism or rules remains in development. Simulating human behaviour, particularly during emergency evacuations, is extremely challenging due to the paucity of robust data and the diversity of individual reactions under extreme conditions. There is consensus that amidst panic, people often attempt to evacuate via their building entry route, ignore nearest exits while searching for others, and follow crowds into unsafe areas [21]. Capturing such heterogeneity in simulation models is crucial for reliably predicting human behaviour, yet remains challenging. Consequently, ABM of pedestrian evacuation is still in a nascent stage of development with AI integration, and elucidating novel methodological frameworks to confront the existing limitations of agent-based approaches can greatly benefit the modelling community.

There exists a limited but substantive corpus of literature reviews utilising ABMs on pedestrian evacuation behavioural modelling. [22] reviewed ABMs of pedestrian evacuation to categorise how social interactions and communication between agents are simulated. [23] conducted a general meta-analysis of 81 papers on ABMs for pedestrian evacuations. Their findings summarise dimensions such as scenarios, software, agent traits, and validation and identify a lack of standardised frameworks for modelling behaviours and validation. However, it remains unclear on challenges in developing ABMs for evacuation modelling and comprehensive discussion on how to overcome issues. [24] provided a high-level overview of different modelling techniques, including ABM, for simulating human evacuation behaviour and identified open questions around how existing models represent behaviours, their implementation, their validation, and their limitations to pick appropriate approaches for evacuation modelling. While existing reviews have surveyed the landscape of evacuation models, gaps remain around the critical analysis of the evacuation modelling methodologies including decision-making models, ABM tools and techniques, behavioural and validation frameworks, behavioural features and delineating major methodological issues and advancement pathways.

This paper presents a systematic literature review of 134 peer-reviewed papers published from 2013 to 2023 that used ABMs to model pedestrian evacuation. Moreover, while previous work has surveyed studies considering simulating pedestrian evacuation, this study focuses more on the methodological challenges of ABM in pedestrian behavioural modelling while suggesting solutions. This timely synthesis promises to guide tool developers, modellers, and policymakers working to translate evacuation simulations into tangible emergency response improvements.

According to the conceptual diagram in Fig. 1, Section 2 presents the methodology followed based on PRISMA guidelines for a comprehensive systematic literature review. Section 3 discusses the main research findings for simulating pedestrian evacuation, outlining behavioural frameworks, agent characteristics, validation techniques, challenges faced by ABMs in evacuation modelling, and relevant solutions. In Section 4, the future research areas according to findings are discussed, while in Section 5, we draw some conclusions based on the surveyed papers and provide direction for future work.

2. Methodology

A systematic Literature Review (SLR) involves a structured process of collecting, understanding, synthesising, and evaluating scientific works to establish a solid theoretical foundation for comprehending a specific topic. Unlike traditional literature reviews, an SLR is a research endeavour that focuses on addressing specific research questions by systematically analysing existing literature, aiming to ensure methodological rigour and minimise biases and distortions in data interpretation [25–28].

2.1. Goal and research questions

The objective of this SLR is to provide an overview of the current research landscape concerning behavioural modelling in an emergency evacuation setting. The study aims to achieve three main goals: (i) describe the current state of research in this field, (ii) identify key research gaps, and (iii) propose potential directions for future investigations. Specifically, this investigation aims to tackle the primary research questions mentioned below.

1. **RQ1:** What cognitive, social, psychological and demographic factors have been considered in modelling individual agents in ABS evacuation models?
2. **RQ2:** What agent architectures and decision-making models are commonly used to represent pedestrian evacuation behaviour?
3. **RQ3:** What ABM software is utilised in existing evacuation simulations?
4. **RQ4:** What approaches are used to validate agent based evacuation models?
5. **RQ5:** What are the challenges faced by ABMs for simulating emergency evacuations?
6. **RQ6:** What are the future research directions to enhance evacuation behaviour modelling?

2.2. Research protocol

To ensure the scientific validity of this review, the Preferred Reporting Items for Systematic Reviews (PRISMA) guidelines [29–31] were allowed. Systematic reviews commonly lack awareness of the established guidelines. However, PRISMA offers a recognised and widely accepted methodology. By employing a checklist of guidelines, PRISMA enhances the quality assurance and replicability of the review process. The subsequent subsections provide a summary of the protocol utilised in this study.

2.3. Information sources and search strategy

The systematic review is based on indexed publications from Scopus Document Search database, which consists of multi-disciplinary publications covering a broader range of scientific resources collections with better metrics than other academic databases [32]. The search for relevant literature on ABMs in evacuation behavioural modelling involved sorting through a large number of articles based on predetermined keywords related to humans, behaviour, evacuation and ABM. To maximise the inclusion of eligible studies, the search terms were expanded with synonyms and modified in conjunction with the use of Boolean operators and the asterisk (*) wildcard character. A search for records within the database was carried out using titles, keywords, and abstracts to achieve a better chance of identifying suitable articles. In the initial keyword search of the academic database, the time span was set to “all years”, and the type of documents was set to “all types”.

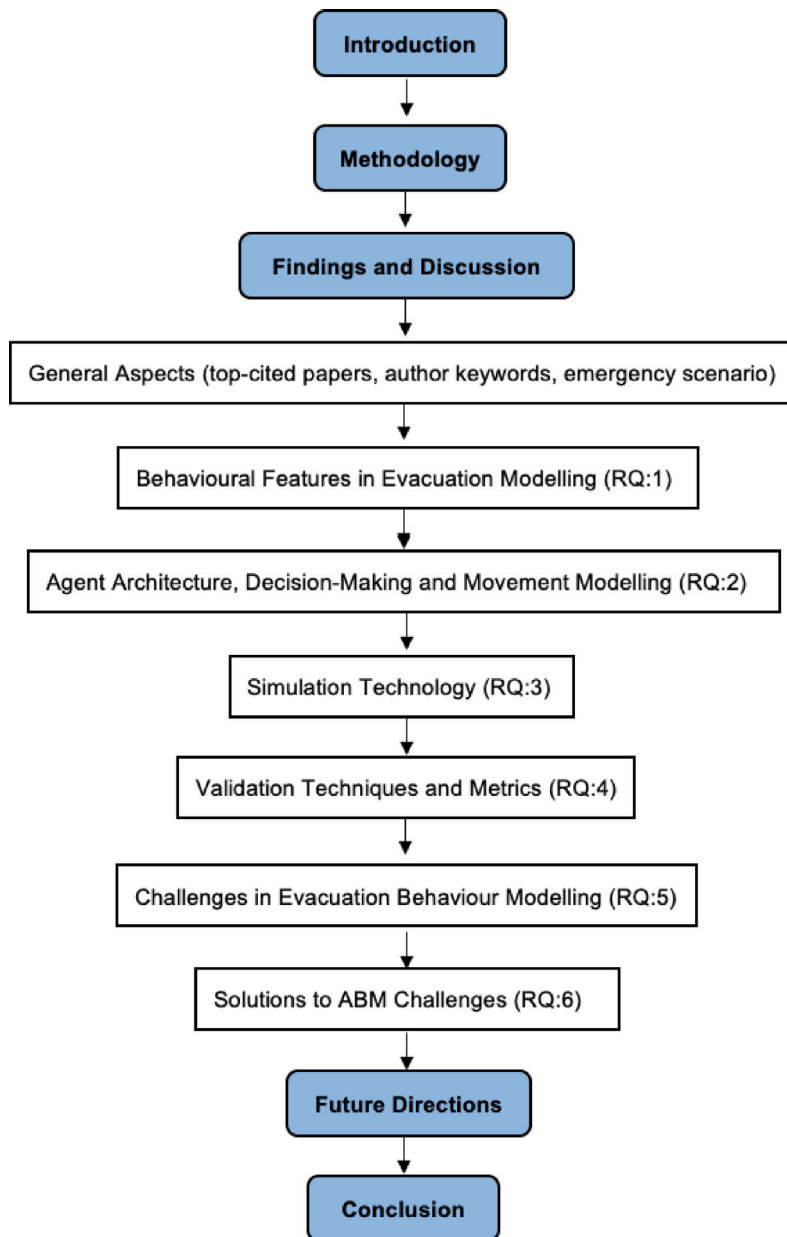


Fig. 1. Conceptual diagram outlining the paper overview.

2.4. Eligibility criteria

Subsequently, the search was further refined to limit the search to databases on the basis of (a) discipline: any discipline relating to Computer Science, Social Science, and Civil and Environmental Engineering; (b) time span: any publication from 2013 until 2023; (c) articles that have been peer-reviewed; (d) language: English. The search result from the database was exported into CSV and BibTeX format. The citation details of all of the retrieved papers were then consolidated into a Microsoft Excel file. Table 1 displays the search outcomes for the search string with and without the database limitations mentioned above.

2.5. Selection of sources

After carrying out the search, the PRISMA guideline was used to identify, select, assess, and refine the final data set of literature used in this study. The researchers conducted an initial rigorous selection based on the title, keywords, and abstract of the articles.

Table 1
Individual search results from Scopus.

Search	Without limitations	With limitations
(Human OR Individual* OR Agent* OR Pedestrian) AND (Behavio* OR Decision Making OR Decision-Making OR Choice*) AND (Emergency OR Evacuation OR Disaster) AND (ABM OR ABS OR Agent Based* OR Agent-Based*)	4,724	500

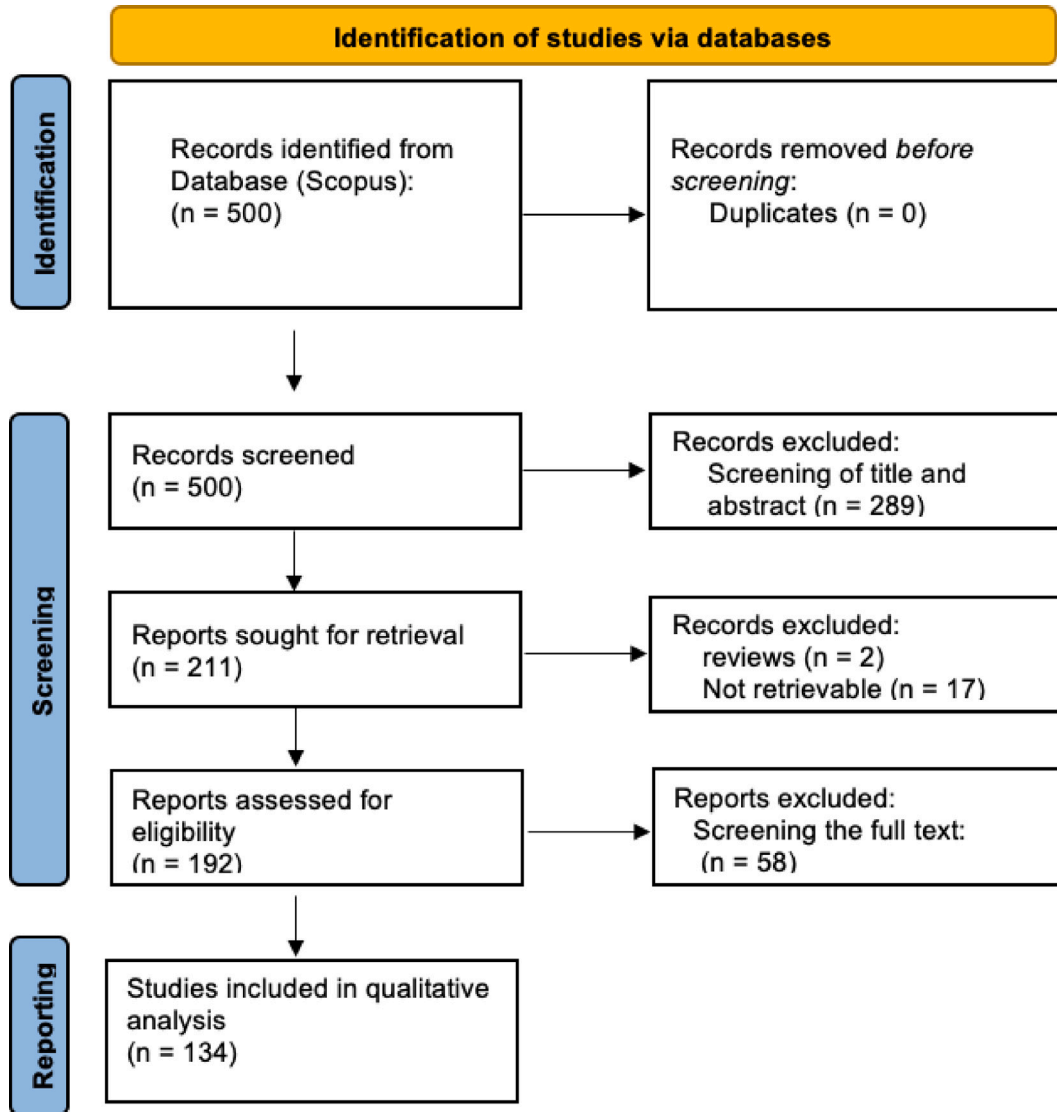


Fig. 2. Search strategy [31].

To be included in the initial selection, the articles had to address the pedestrian behavioural modelling in ABMs in the context of emergency evacuation scenarios. In cases where the title or abstract did not provide sufficient information, the paper was selected for further examination. The reviewers engaged in discussions and **deliberations** to arrive at a **consolidated** list of publications. Fig. 2 shows the overall research strategy used for paper selection in the SLR. After final screening, the papers being considered for inclusion underwent further analysis employing the data charting method outlined in the subsequent subsection. The final set of articles for the review was then used for statistical analysis using VOSviewer and general reference purposes.

The review was conducted based on 134 journal papers from the Scopus database, and a multi-stage screening process was applied to the results. First, 289 records were excluded on the basis that neither the title nor the abstract mentioned pedestrian evacuation

modelling in ABM, for example, papers investigating medical emergency research such as antibiotic stewardship, contraception, and COVID-19 management, non-human behaviour such as voltage instability in power systems, wireless stethoscopes, and unmanned aerial vehicle (UAV) inspections, and non-emergency situations such as knowledge representation and sharing in risk management. In addition, papers related to traffic simulation in an emergency, disaster response after an emergency and terrorist evacuations were also excluded. In the second screening stage, 19 records were removed because they were either review papers or irretrievable. Thirdly, 58 articles were removed after careful reading and having identified them as not being focused on the specific ABM applications of interest, namely disaster response, resource or task allocation, formation of emergency relief alliances (post-evacuation) or risk communication (pre-evacuation). This resulted in the identification of 134 suitable articles suitable for inclusion in the SLR.

2.6. Data charting

The authors collaboratively designed a data charting form to identify the variables to be extracted. Subsequently, the data were individually charted, the authors engaged in discussions to analyse the results. For every paper that was retained following the screening, the subsequent data elements were gathered:

- Author, title, year, the paper cited count and author keywords of publication: as indicated in the citation provided by the database.
- Aim and objectives
- Approaches used to validate the implemented ABS evacuation models
- Specific ABS tools, algorithms, and programming languages used
- Agent architectures/theoretical frameworks and decision-making models used
- Factors considered in the modelling of individual agents
- Challenges presented in the ABM approach in simulating pedestrian behaviour
- The study's main findings and limitations

2.7. Synthesis of results

After the final screening, an analysis of the overall results was conducted to provide an overview of the existing literature on ABM in pedestrian evacuation modelling. Subsequently, the focus shifted to ABM methodologies, allowing us to examine the breadth (challenges of ABM and corresponding solutions) and depth (level of maturity) of the existing research. The distinctive attributes of each publication included are presented in tabular format, with the frequencies charted data and depicted in graphical format. Finally, the findings of the review for each of the identified research questions were summarised and discussed.

2.8. Strengths and limitations

To ensure the inclusiveness of our review, the PRISMA methodology was employed, and discrepancies were addressed through thorough discussions among authors. While aiming for an international perspective, our search was limited to one reputable database known for the quality of its contributions to scientific research, ensuring the rigour and quality of the studies included in our review. Limitations were introduced on the basis of the restrictions on the dates of publication imposed, the language (English only), and in the area of research.

3. Findings and discussion

The papers included in this review deliberate the development of ABMs for pedestrian evacuation, are peer-reviewed, and were published within the last 10 years (2013–2023). Fig. 3 illustrates the number of papers included in the SLR over this time period. It depicts that 64% of the papers were published after 2017; between 2013 and 2017, the average number of publications was nine a year, while between 2017 and 2023, the average was 16, suggesting an increase in interest in the topic over time.

3.1. General aspects of papers related to pedestrian evacuation modelling

3.1.1. Top-cited papers and high impact journals

Table 2 and Fig. 4 illustrate the ten top-cited papers related to Pedestrian evacuation modelling in ABMs and their associated journals, respectively. The papers and the respective journals are arranged in chronological order in terms of citation count. The journals "Safety Science" [33,34] and "International Journal of Disaster Risk Reduction" [35,36] dominate in terms of impactful research, each accounting for two out of the top ten most influential papers. Experts recognise *Safety Science for its multidisciplinary nature of public safety*, covering emergency response, resilience infrastructure, and community preparedness and recovery, while the *International Journal of Disaster Risk Reduction focuses specifically on integrative research and policy guidelines for building disaster resilience worldwide*. However, based on the citation per year values in Table 2, the papers from Information Sciences and Transportation Research Part C: Emerging Technologies have achieved the highest rates. The significance of these top-cited papers is that they capture key behavioural factors, spatial complexity, and dynamic changes in the environment. For example, [37–39]

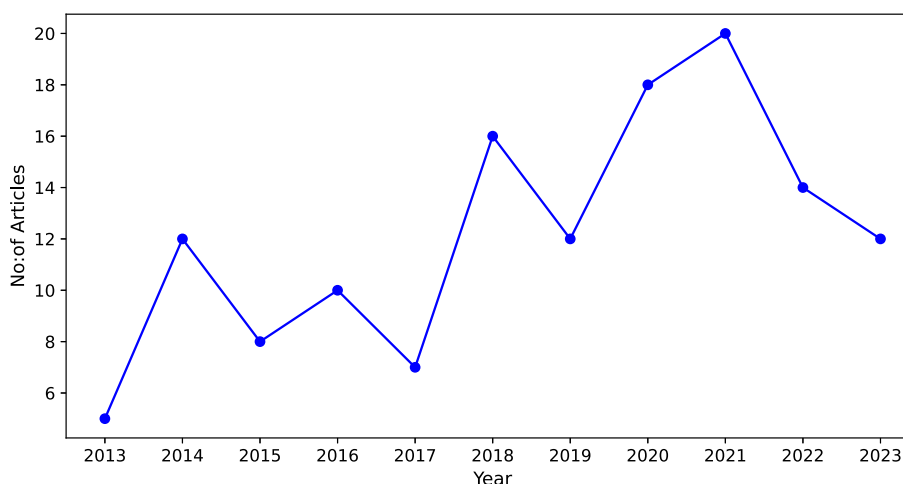


Fig. 3. Number of surveyed articles per year.

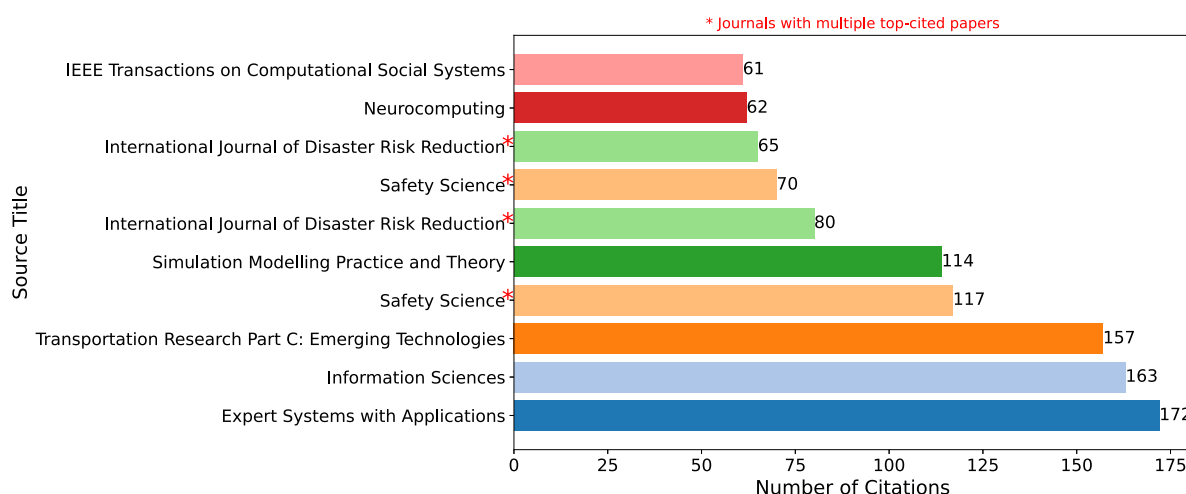


Fig. 4. High impact journals based on top-cited papers. Journals with the same colour and red star (*) denotes journals with multiple top-cited papers (Safety Science and International Journal of Disaster Risk Reduction).

specifically focused on integrating psychological and cognitive theories for decision-making, panic, grouping, and communication into their models, while [38,40] demonstrated the importance of representing the environment realistically incorporating obstacles, fire spread, and activation of safety features. Moreover, proposed hybrid modelling frameworks that combine ABM with other techniques such as cellular automata [41], social force models [34], or affordance theory [39] created more comprehensive and realistic simulations. Some applied the developed ABM frameworks in practical emergency planning through case study simulations aiming to evaluate emergency preparedness strategies, identify bottlenecks, and test the impact of various behavioural or environmental parameters on evacuation efficiency [35,36,42]. While validation remains a challenge, several papers attempt to incorporate empirical data or conduct qualitative validation of their models. [33] used video data of real evacuations to inform their behavioural model. [34] validated their model against a real-world evacuation drill.

3.1.2. Analysis on author keywords

Fig. 5 presents a visual network mapping, using VOSviewer, of author keyword occurrences from the collected papers, excluding common top terms such as “agent-based modelling” and “evacuation”, which were overwhelmingly prevalent. It identifies several notable clusters. The red cluster centres around psychological and cognitive aspects of evacuation decision-making using game theory and Bayesian Nash equilibrium [43,44], highlighting a research priority on incorporating realistic human behaviour. This is surrounded by several small clusters (yellow, purple, ash), highlighting the connection with several decision-making factors such as route choice, egress and panic. Another tightly interlinked green cluster indicates earthquake evacuation with a substantial link on using “virtual reality” [45] and “fds+evac” [46] to represent space and simulate movement. Further, terms such as “social force model” and “discrete choice modelling” reveal the hybrid ABM methodologies that bring dynamic movement patterns considering

Table 2

Top 10 cited papers. The star (*) denotes journals with highest citation per year.

Year	Reference	Source title	Objective	Citations per year
2014	[40]	Expert Systems with Applications	Evaluate emergency preparedness protocols during fire in concert venue	19.11
2015	[38]	Information Sciences	Evaluate the impact of occupant adaptivity with spatio-temporal changes in fire safety systems.	20.38 *
2016	[42]	Transportation Research Part C: Emerging Technologies	Investigate the impacts of evacuee decision-making time and provision of vertical evacuation structures.	22.43 *
2014	[33]	Safety Science	Define the chronological sequence of behaviours during an earthquake evacuation.	13.0
2013	[39]	Simulation Modelling Practice and Theory	Simulate human behaviour using affordance-based approach.	11.4
2017	[35]	International journal of disaster risk reduction	Investigate the impact of visitors on the evacuation process during tsunami.	13.33
2019	[34]	Safety Science	Improve evacuation planning for a multi-floor university building.	17.5
2019	[36]	International Journal of Disaster Risk Reduction	Assess the impact of shelter locations on the effectiveness of vertical evacuation during tsunami.	16.25
2014	[41]	Neurocomputing	Analyse complex pedestrian decision-making dynamics in complex environments like stadiums or shopping centres.	6.89
2018	[37]	IEEE Transactions on Computational Social Systems	Identify environmental bottlenecks and evaluate evacuation strategies under panic situations.	12.2

repulsive and attractive forces between agents, obstacles, and destinations while capturing the heterogeneity in decision-making among agents based on their individual preferences, attributes, and the available alternatives [47–49].

3.1.3. Emergency type and the environment considered

The type of emergency and the environment highly influence the behavioural factors and model features being simulated. Fig. 6 reveals that most of the studies did not target a specific type of emergency evacuation (labelled as “Emergency evacuation” in Fig. 6) [43,50–52]. From the rest that specified an emergency, there is a skew towards fire or smoke on the evacuation process. Fire location, heat release rate, smoke spread, and staff actions are shown to be key factors affecting evacuation routes, times, visibility, and occupant survival [38,53–56]. Integration of detailed fire and smoke dynamics with ABM has been achieved via methods such as the Fire Dynamics Simulator (FDS). This can model combustion, radiation, pyrolysis, and smoke density [46,57]. Further, [58] extended ABM called FREEgress incorporating heat, temperature, and toxicity.

Other studies modelled the evacuation of cities during tsunamis, earthquakes, wildfires, Hurricanes and floods. It is important to recognise that each scenario presents unique challenges that necessitate a tailored response. For instance, the case of a tsunami evacuation; one critical factor that must be taken into account is identifying the optimal locations for vertical evacuation structures to ensure the safety of those seeking refuge [36]. During a tsunami, horizontal evacuations can become complicated, especially with traffic and damage to transportation infrastructure during the earthquake. Rapidly rising water levels can pose significant challenges, not only to the process of evacuating people horizontally to safer areas, but also to the structural integrity and safety of vertical evacuation shelters [35,42,59,60]. Another major variance between different types of emergency evacuations is the occurrence time, length, and time required to evacuate. For instance, near-field tsunamis arrive within several minutes after the initial earthquake, expanding over several kilometres of land, and tsunami during nighttime makes pedestrians more vulnerable [61,62]. In the case of an earthquake, evacuees may seek shelter under a piece of furniture within a few seconds of feeling the shaking of the building [63,64]. In contrast, when it comes to disasters such as floods, hurricanes, and wildfires, those needing to evacuate typically receive warnings hours before the event, providing them with more time to prepare. In these situations, evacuees commonly use vehicles as their primary means of transportation to reach safe shelter [13,65–67]. The Fig. 6 shows a growing interest among researchers in recent years in examining pedestrian behaviour in large-scale city-level emergencies [68,69].

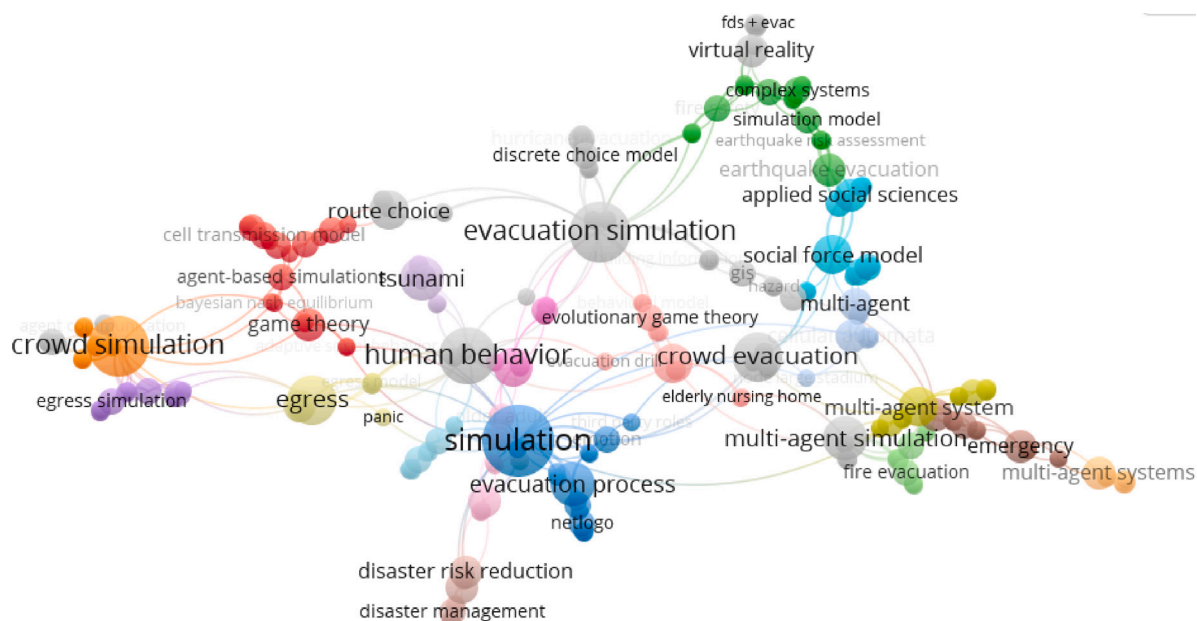


Fig. 5. A bibliometric network of keywords to pedestrian behaviour during an emergency. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

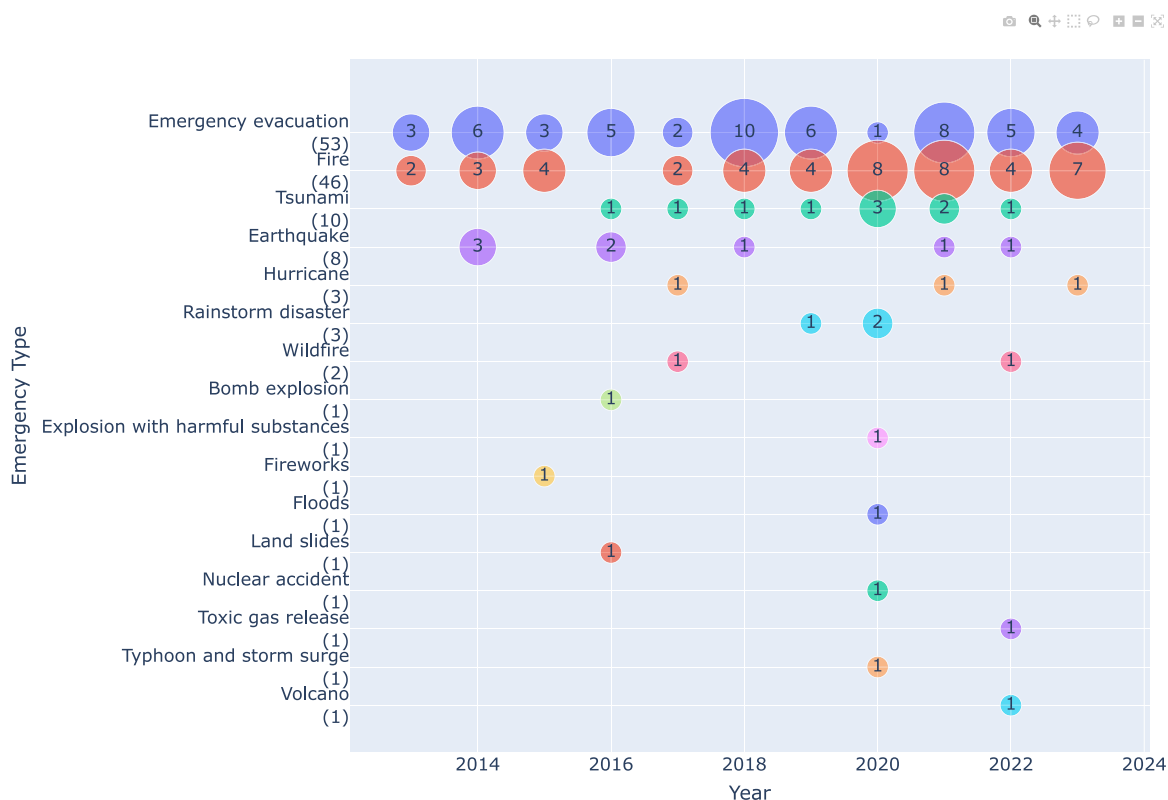


Fig. 6. Emergency types over the years.

The majority of models underwent testing in particular settings, and their applicability is constrained to certain categories of structures due to inherent limitations. The targeted case studies fell into different environmental categories as shown in Fig. 7: any

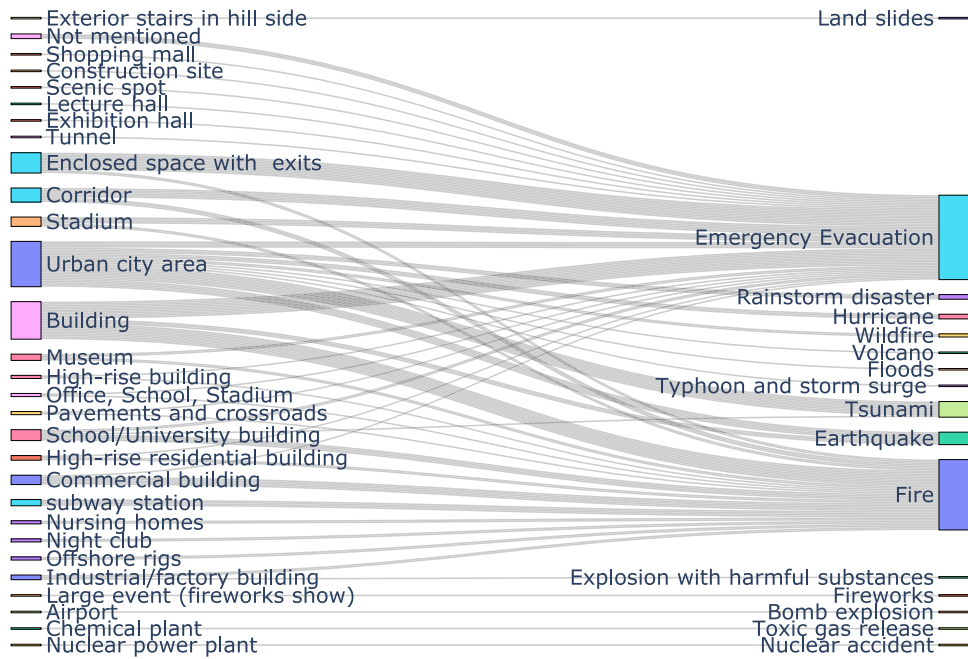


Fig. 7. Relationship between emergency type and evacuation space.

building [70,71], the urban city [72,73], high-rise residential buildings [14,74], commercial building [15,75], public transportation stations [76,77], university/school buildings [55,78], enclosed spaces with exits [16,79]. Other categories include corridors [48,53], and offshore rigs [80]. This suggests the models may have been tailored to the unique characteristics and requirements of certain evacuation contexts. The lack of natural light and air circulation in subterranean spaces such as subway systems can significantly impair pedestrians' spatial awareness and sense of direction. This heightened risk of suffocation due to limited ventilation poses substantial hazards to those seeking shelter or evacuation routes in these underground environments [44,81]. Moreover, the moving dynamics of pedestrians differ with the building familiarity. For example, individuals in museums, stadiums, exhibition halls and metro stations are usually gathered near ticket offices and vendors [40,50,82], while occupants of a residence or an office building are usually familiar with the building layout [14,83]. The physical characteristics of a building are strongly related to the human decision making process [84]. For example, the capacity of a building impacts pedestrians' proximity and congestion during the evacuations. Proper allocation of wider doors, multiple exits, fire shutters, and exit signs can improve evacuation efficiency [38,57,58,71,85–88]. The navigation of agents is guided by exit doors and signs, which serve as focal points providing directional cues. Every element within the environment possesses unique attributes, including its spatial position, shape, and size. In an urban evacuation context, the landscape consists of various components such as roadways, structures, overpasses, and barriers. Specifically, the models account for the locations of shelters and safe areas, accessibility and proximity of these sites, and the condition of transportation options like vehicles and roads [89,90]. Factors such as the presence of debris, floodwaters, or other environmental obstructions can impact mobility [63]. This recommends the incorporation of a proper dynamic route planning system. The model by [91] showed positive results using the Evacuation Route Multi-Agent Recommender System (ERMARSys) for updating paths based on sensor data.

In summary, the influential papers tend to focus on capturing realistic human behaviours, representing complex environments, and proposing hybrid modelling approaches. Author keywords highlight clusters around psychological factors, earthquake evacuations, the integration of techniques such as virtual reality and hybrid ABMs with discrete choice modelling. While many studies do not target a specific emergency type, there is a notable emphasis on fire evacuations and a growing interest in large-scale city-level emergencies. The models are often tailored to specific environments, such as high-rise buildings, public transportation stations, and urban areas, each presenting unique challenges and considerations. As research progresses, there is an increasing focus on modelling evacuation protocols across wider geographic areas and transportation networks beyond just building-level emergencies. This trend reflects the need for emergency planners to contend with emergent public conditions in complex, uncontrolled settings. The remaining outcomes of the systematic review are organised according to the research questions outlined in Section 2.

3.2. Behavioural features in evacuation modelling

Human characteristics, behaviour and environmental factors highly impact the evacuation process for creating more realistic simulations [46]. Humans in real life have different psychological needs and physiological properties. Therefore, it is advantageous

to model evacuating crowds as heterogeneous individuals rather than homogeneous [92]. Hence, identifying all the characteristics affecting evacuation behaviour is a critical stage in model development. The analysis of Fig. 8 provides insights into the common behavioural attributes in pedestrian evacuation modelling, with fundamental factors such as crowd density, collision avoidance, speed, and route choice.

3.2.1. Physiological and demographic factors

When considering the internal state and external environment, there are several factors that are relevant to evacuation that need to be incorporated into evacuation model development. The internal state includes physiological, psychological and demographic factors, while interactions among the two or more internal states of agents affect sociological factors. Together with environmental factors, an agent system emerges. All these characteristics influence pedestrian behaviour during an evacuation in regard to decision-making and movement. Demographics played an important role in the evacuees' movement. Age and gender impact movement [54,88,93], for example, [68] grouped individuals into several speed categories (men, women, children, older people) while also considering the shoulder width. On the other hand, [51] assigned speed to agents where walking speed increases until age 25, constant between 25–60, and then declines as people get older, with men walking faster than females. Moreover, health and disability affect mobility and decision-making [92,94,95]. Physical strength and stamina may change according to age and gender, and endurance capability may be lower for people with disabilities after the emergency. Hence, physical characteristics heavily affect an individual's choice of action, such as which exit to choose based on the distance or free of obstacles. [86] reduced speed for the agents using a wheelchair, while [96] considered the impact on injuries during evacuation for speed reduction and seriously injured agents needing rescue by ambulances. From the studies conducted in nursing homes, it is evident that the elderly with disabilities, including mobility levels, hearing, and vision impairments, require some sort of assistance during evacuation. These models need to simulate individual and group behaviours with interactions between caregivers and elderly agents [94]. Similarly, assigning different personalities (consciousness, agreeableness, neurotic and extraverted personalities) mostly following OCEAN model [83,97,98] able to simulate a diverse population with attributes sampled from appropriate distributions.

3.2.2. Psychological factors

A variety of psychological and cognitive factors influence individual behaviours during emergencies. Agents would not always select the shortest route; instead, they would evacuate through the most familiar route. Factors such as affinity and conformity can lead agents to blindly follow the majority's action. On the other hand, agents in the heightened stress state can exhibit unsafe actions like pushing, shoving, and not following instructions, exacerbating the evacuation difficulties [14]. Some studies used a "panic coefficient" affecting movement speed and herd mentality; thereby, they have shown that panic is one of the main causes of casualties [54,97]. Consequently, many simulations considered the panic and stress aspects during evacuation. For example, in [14], the panic was measured according to two different factors: distance to the hazard source and the perceived crowd density. In [97], psychological stress was calculated by considering their personality traits of neuroticism, conscientiousness, extraversion, and agreeableness, which determine their levels of panic and impatience. Several studies evaluated perceived risk levels (risk-seeking, risk-averse, risk-neutral, best-response) in response to hazard cues [68,89,99] and emotions such as fear affecting movement [56,100] as sources to determine panic. Similarly, Learning and adaptivity was incorporated in [101] model, with agents able to update knowledge and adjust behaviour based on experiences over time. Familiarity with building layouts and exit locations also plays a role, with agents designated as residents having more awareness than visitors [38].

3.2.3. Sociological factors

Another important aspect during evacuation is human relationships and interactions. For instance, families and tour groups are evacuating together and interacting within the group [102]. This may show seeking behaviour where one member is searching for missing family members [89,102]. Even though many surveyed papers assume that all pedestrians possess the same level of knowledge about the environment, some allow the laymen to follow instructions from leaders [83,103,104]. [77,82] found that close group gathering with leader following behaviour has much impact on directing agents to the right exit, especially visitors. These social factors drive important collective behaviour through local interactions between agents and within groups. However, there are complex socio-psychological dynamics that play during evacuations involving social categorisation, identification, trust, and credibility that shape decision-making. This may include social influence, information-seeking and sharing, and information distortion between agents [49,68,80,104,105], and helping behaviours with agents choosing to assist others in need [70,72]. Herd mentality is also included in some studies [66,97] with agents demonstrating conformity by following crowd movements when panic levels rise past the threshold. Additionally, social force is another factor that mainly handles the movement of evacuees while emerging crowd level dynamics. It includes both attraction and repulsive forces. Agent's movement direction may be guided by the attraction forces to safety signs, exits, etc. [12], while repulsive forces avoid obstacles, walls, and injuries [106]. [102] modelled relationships such as family and friends using attraction forces to meet first before evacuating together.

3.2.4. Environmental factors

According to affordance theory, certain actions during evacuations are based on the pedestrian's perception of the environment [107]. As identified in the surveyed studies, external environmental factors during evacuation can be categorised as outcomes of the hazard, physical characteristics of the environment, and outcomes from agent behaviour. For example, earthquake intensities, hurricane strength, visibility and smoke levels during a fire impact on velocity and breathing difficulties [62,65,79,80,103]. Similarly, the damages to the infrastructure and the presence of debris and rubble are affecting the availability of exits and

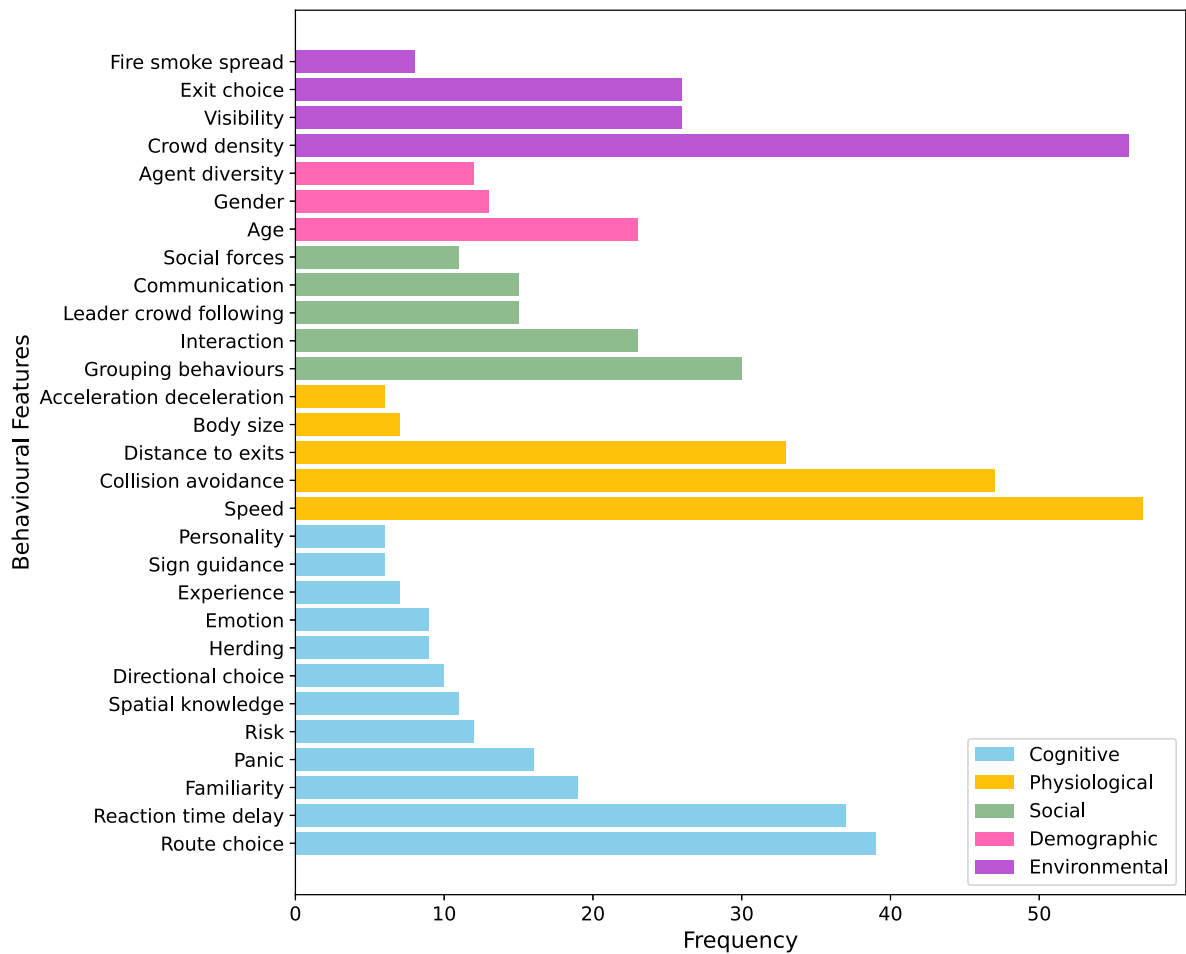


Fig. 8. Top 30 most common behavioural features in evacuation modelling.

navigation [59,63,73]. All the physical characteristics of the buildings and terrain factors, such as slopes, staircases, exit doors and signs, etc. are incorporated into the models [108]. Random obstacles are also generated in some models to evaluate their effect on congestion and evacuation times [109]. The outcomes of the agent behaviour are overtaking slower agents, crowd density at exits, and relative factors from neighbouring agents such as relative velocity. More realistic evacuation simulations could integrate complex, data-driven models of dynamic environments (e.g. weather, terrain, infrastructure damage) coupled with behavioural models to capture emergent human–environment interactions.

In summary, ABMs for pedestrian evacuation have incorporated a wide range of physiological, psychological, sociological, and environmental factors to capture realistic human behaviour and decision-making processes. Key physiological and demographic attributes include age, gender, health status, and disabilities, which impact movement speeds and route choices. Psychological factors such as panic, stress, learning, risk perception, and emotions are often modelled to represent the mental states of agents during emergencies. Social interactions, group dynamics, information sharing, and helping behaviours add further complexity, with agents exhibiting phenomena like leader following, herd mentality, and social influence. **Environmental factors, including hazard characteristics, physical infrastructure, and dynamic conditions, shape the context in which agents navigate and make decisions.** While these models have made significant strides in representing human behaviour, challenges remain in validating the distributions of agent traits against real-world population data and quantitatively linking environmental perceptions to observable actions. Future work integrating data-driven environmental models with behavioural models could enable more realistic simulations of emergent human–environment interactions during evacuations.

3.3. Agent architecture, decision-making, and movement modelling

The basis for the model development may be well-defined behavioural theories or empirical data, if not modeller expertise and assumptions [110,111]. However, collecting actual data and analysing individual reasoning from a video recording of a real dangerous scenario is always challenging. Even the data collected through experiments such as drills and VR would not

provide accurate behavioural patterns from participants, knowing there is no real danger [23]. Therefore, following an established framework as the basis for behavioural factors integration to the micro-level agents is essential [112]. This raises the need to select the right variables and computational powers to scale complex agents to simulate larger crowd dynamics. Comprehensive agent architectures/behavioural theories and advancement in computer performance may overcome these issues.

3.3.1. Agent architectures and behavioural frameworks

The proper design of agent architecture is crucial in building intelligent agent systems. The agent can be reactive, deliberative, or hybrid, which integrates the advantages of the two to achieve a goal-oriented behaviour by decomposing the plan into several sub-plans [113]. Most of the reviewed papers followed custom agent architectures. Some used a distributed knowledge base of if-then rules that agents use to interpret their environment, activate relevant rules, execute actions, and update their internal state [58,106,114]. Others integrate many established theories and frameworks together to design their deliberative agents. The BEN (Behaviour with Emotions and Norms) architecture is a modular agent architecture that uses cognition based on the BDI (Belief-Desire-Intention), emotions based on the OCC (Ortony, Clore, and Collins) theory, personality based on the OCEAN (Openness, Conscientiousness, Extroversion, Agreeableness, Neuroticism) model, emotional contagion based on the ASCRIBE (Agent-based Social Contagion Regarding Intentions, Beliefs and Emotions) model, social relations based on Svennevig's framework, and norms and obligations based on the BOID (Belief, Obligation, Intention, Desire) architecture, allowing modellers to create expressive and credible agent behaviours for social simulations [98]. Similarly, Behaviour Oriented Design (BOD) agents in the research by [78] are structured around reactive behaviours that can be composed hierarchically to produce more complex decision-making. In comparison, BDI may be more appropriate for agents operating in knowledge-rich environments that require more deliberative reasoning [115].

In some cases, researchers utilise behavioural frameworks that describe human behaviour and decision-making processes during emergency situations. The Protective Action Decision Model (PADM) is one such conceptual model. The PADM consists of several stages, which include pre-decisional processes, assessing the situation and perceiving risk, searching for and selecting protective actions, and implementing the chosen protective action. The research by [54] focuses on the final stage of the model (protective action implementation) to incorporate the PADM into the ABM by considering how agents' individual characteristics, interactions, and interpretations of environmental cues influence their pre-evacuation time and ultimate decision. Similarly, [116] proposed the Dynamic Recognition-Primed Decision (D-RPD) model as a core behavioural framework and decision-making architecture for the individual agents, with the agent attributes and decision rules designed to capture the bounded rationality and adaptive risk preferences of humans during an urban flood evacuation scenario. Here, the classical Recognition-Primed Decision (RPD) model is extended to incorporate dynamic updates to the agents' experience base.

The currently evolving, research behavioural frameworks are mostly Machine Learning (ML) based. The NEAT (NeuroEvolution of Augmenting Topologies) based ABM architecture presented in this study evolves artificial neural networks (ANNs) to control the decision-making and movement of individual agents in a crowd simulation. Each agent has a set of sensors that provide input to the agent's unique ANN controller. The ANN processes these inputs to generate outputs that determine the agent's velocity and direction of movement at each time step, with the ANN's structure and weights optimised over generations using a Genetic Algorithm (GA). The evolving process ultimately leads to the evolution of complex, optimised ANNs that can effectively control the agents' behaviour in the simulation. However, what is lacking in this methodology is the interpretability of agent evacuation behaviour and accuracy of the data used for the process [101].

3.3.2. Decision making models

Analysis of the identified literature on human evacuation behaviour modelling reveals a variety of modelling approaches adopted for representing the decision-making processes of agents (see Fig. 9). The most common technique employed across the studies is to use pre-define rules ranging from simple deterministic to stochastic rules [14,70,117,118]. These rules, such as exit choices, route selection, and movement speeds are encoded based on expert knowledge, behavioural theories and architectures such as BDI [115], and rarely based on data collected through experiments and surveys [17,100]. The strong assumptions and simplifications may fail to capture the full complexity of human behaviour, while using more intricate cognitive models and accurate evacuation data would provide the most precise and understandable results [23].

Several mathematical methods have been applied to encode precise rule structures for simulating cognitive processes. The rules with utility maximisation provide more flexibility; however, they remain constrained by the need to define utility function [50,98]. The utility function may differ according to the factors considered and the decision-making scope. For example, the affordance-based model focuses on maximising utility based on the immediate local spatial factors of neighbouring cells [50], while the spatio-temporal potential field model takes a more strategic and proactive approach by considering the global spatio-temporal cost of entire routes to the exit [76]. In a game-theoretic approach, the utility functions of agents are interdependent, implying that an individual's outcome is contingent not solely on their personal decisions but also on the choices made by other players. For example, in a crowd evacuation scenario, agents strategically choose an exit that maximises their expected payoffs, taking into account the anticipated actions of other agents in order to avoid congestion or competition for limited space [57,99]. It facilitates modelling strategic thinking and multi-agent interactions; however, it may not effectively emulate bounded rationality under stressful conditions [119]. As a solution, [43] combined Bayesian Nash Equilibrium with game theory, enabling agents to maintain and update personal probabilistic beliefs about unknown aspects of the game. It enables agents to intelligently avoid congestion and evacuate more quickly by predicting neighbour occupancy.

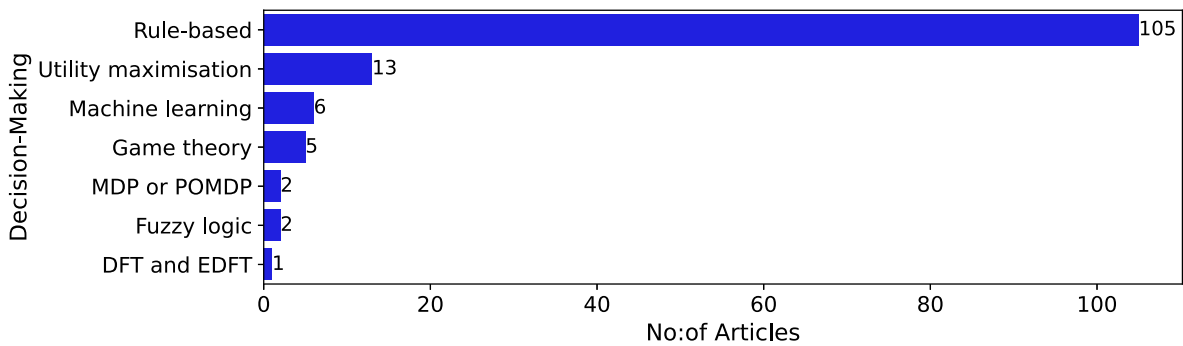


Fig. 9. Number of surveyed articles per decision-making methodology.

Fuzzy logic is another mathematical framework used for decision making in evacuation modelling for reasoning with uncertain or imprecise information. These rules map input variables (distance, crowd density) to output variables (desirability of a route) through membership functions. These membership functions use if-then rules to define the degree to which a value belongs to a fuzzy set [120]. Among the review articles, [54] used a fuzzy inference system to model uncertainty in an occupant's diagnosis of a situation, while [120] proposed using fuzzy logic to simulate panic behaviour and generate crowd movement predictions. In some cases, researchers use Markov Decision Processes (MPD) and Partially Observable Markov Decision Processes (POMDPs) for sequential decision problems under uncertainty. In this approach, the agent makes decisions based on the current state (in MDP) or belief state (in POMDP) and the expected future rewards. The optimal policy is often learned through techniques such as dynamic programming or Reinforcement Learning (RL) [72,92]. The paper by [65] employs the Decision Field Theory (DFT) and its extension called Extended Decision Field Theory (EDFT) frameworks from the field of mathematical psychology to model the dynamic, stochastic evolution of preferences that drive agent evacuation decisions, while accounting for environmental factors and social interactions. The approach aims to realistically capture heterogeneity and stochasticity in human decision making under evolving hurricane disaster scenarios.

The use of ML in pedestrian evacuation modelling is limited but emerging. This could help infer patterns and complex relationships from data. This is likely to expand with the availability of richer data sets on human evacuation dynamics, often from evacuation experiments. The integration of ML techniques to encode agent behaviours and decisions is an emerging capability that has been leveraged by a subset of studies. For instance, [12] applied SARSA algorithm with RL for experience sharing to allow agents to dynamically select efficient evacuation routes and exits while avoiding crowding by propagating learned policies. [16] adopted deep RL to develop policies and navigate agents based on a human-like 3D perceptron, enabling them to learn complex behaviours, such as congestion avoidance through experience. In addition, ML has also been adopted for choice modelling. For example, [47] compared ML and discrete choice models to predict way-finding decisions based on factors such as visual access and crowd density. To capture heterogeneity, [69] built a random forest model to model the exit choices of agents during nuclear plant evacuation based on survey data. Overall, the integration of modern ML techniques demonstrates enhanced flexibility in capturing complex human behaviours and decisions during emergency evacuations. The reliability of the model depends on how closely the evacuation data used in the model reflects real-world scenarios and conditions.

3.3.3. Movement modelling

According to Tang and Bennett [121], the main components that influence animal movement through space are the internal state, external environment, and navigational and motion capabilities. Given an individual's dynamic environment and internal state during a disaster, ABMs can capture the two critical decisions: when to move and where to move. Contrary to the normal behavioural condition, in a disastrous condition, people begin to evacuate with a warning alarm and continue it until they reach safety. However, the damages to the environment may result in changes to the internal agent goals, movement decisions, and subsequent movement behaviour [122].

Cellular Automata (CA) is used in conjunction with ABM to model the spatial context in which the agents operate. It consists of a grid of cells, where each cell can be in one of a finite number of states. The state of each cell evolves over time based on certain transition rules that depend on the states of the neighbouring cells. The agent moves to the adjacent cell while maximising the utility based on predefined decision making rules. It is the most commonly used movement methodology for building case studies for its computational efficiency, but lack realism [15,34,40,50,70,114]. In some cases, the direction in which an agent moves in the grid at each step is determined probabilistically using a Markov chain. The probabilities in the Markov chain used by [123] depended on factors such as the agent's current location, the occupancy of neighbouring cells, and the location of the nearest target. In some models, a grid-based environment is used in urban city space. [124] represented urban city space as a grid of cells where agents decide on their velocity on crowd density and direction based on neighbouring cell information. However, in most simulations, the space of urban areas is created based on link-node network approach [51,60,61]

Other studies have explored models that use continuous space for agent movement. The agent velocity and the direction are based on the pre-set values or pre-define rules that consider the surrounding factors to derive the parameters [36,98,125,126]. For

example, [86] derived the agent velocity from visibility and collision avoidance (Reciprocal Velocity Obstacle (RVO) model) and determined the direction based on shortest paths and smoke avoidance. Some studies use the A* path-finding algorithm to find the shortest route that likely sets the preferred direction [127]. Force-based equations of motion have been used in social force models to define more fluid and natural movements, similar to how pedestrians navigate in real life [63,82,128]. The goal is the main attractive force driving agents to their destinations, while neighbour and collision forces provide repulsion to avoid crowding and collisions. Cohesion adds attraction between agents with similar goals. The other forces handle interactions in specific situations like queuing and cornering. The net effect is crowd movement emerging from these attractive and repulsive forces acting on individual agents [129].

Potential field analogies encoded psychological/social forces for agent velocity (attractive and repulsive forces), supporting emergent movements akin to human crowds. However, the calibration and validation of parameters were found to be difficult [79]. Route/path-following models enabled goal-driven navigation by planning evacuation paths dynamically or based on assumed knowledge. However, they did not address factors such as panic or impatience well [81]. Velocity-based models, such as Self-organised Gradient Emergency Models (SGEM), directly manipulate agent speeds at each step based on personal or environmental attributes. [12] used SGEM models to determine the velocity on the basis of crowd density, ranging from 1.4 m/s at low densities to 0.1 m/s at high densities, while collision avoidance with other agents and obstacles was handled through the Optimal Reciprocal Collision Avoidance (ORCA) model, which avoids collisions by adjusting agent velocities. Similarly, [130] updated agent velocities based on the chaos level calculated by boltzmann entropy concept.

In summary, agent architecture, decision-making, and movement modelling are critical components in developing realistic ABMs for pedestrian evacuation. While custom architectures and rule-based decision models are commonly employed, there is a growing trend towards integrating established behavioural theories, cognitive frameworks, and mathematical methods to enhance the realism and flexibility of agent cognition and choice. Machine learning techniques are also emerging as powerful tools for inferring complex behaviours and decisions from data, although their application in evacuation modelling remains limited. For movement modelling, grid-based rule based navigation approaches dominate due to their simplicity and computational efficiency, while continuous space models offer finer control over agent navigation and steering. However, the incorporation of realistic human mobility constraints and validation against empirical evacuation data remain key challenges. Future research should focus on developing hybrid models that leverage the strengths of different approaches while balancing behavioural accuracy and computational efficiency.

3.4. Simulation technology

The commercially available software utilised to model evacuation behaviour includes NetLogo [13,15,36,43,70,76,93,131], AnyLogic [14,17,54,97,132], Pathfinder [77,85,86,125], Artisoc software [35,60,61,133], Unity [16,96,134], SAFEgress [104,135], Mason [37,78,89], and GAMA [69,98,136]. NetLogo and AnyLogic are the most frequently referenced software in the surveyed papers, with NetLogo utilised in 13.8% of the simulations and AnyLogic in 7.5%, as shown in Fig. 10. NetLogo is a user-friendly ABM environment with an Integrated Development Environment (IDE) for model development, visualisation, and analysis, making it accessible for users without extensive programming experience [15]. AnyLogic provides a comprehensive platform to develop a customised agent-based evacuation simulation by leveraging its built-in pedestrian dynamics and psychological model libraries [94]. Pathfinder is an agent-based egress and human movement simulator that uses a combination of steering behaviours and physical constraints to simulate large-scale behaviour [77]. Unity provides the underlying 3D simulation software and GIS integration for crowd simulation model [83]. SAFEgress integrates with other modelling tools and systems, such as fire dynamics simulators (FDS) and building information modelling (BIM) software [104]. MASON is a Java based simulation toolkit with GeoMASON library for geospatial extension [89]. GAMA is a simulation platform for building spatially explicit agent-based simulations. It can implement different behavioural states and transitions identified from the data. GAMA is implemented in Java but provides a dedicated modelling language (GAML) for coding models [136].

Other simulation software include Repast [66,124] which is written in c++ ideal for large-scale simulations that require parallel computing, MassMotion [117,132], Pedestrian Dynamics [137], EgressSFM that uses the Scalar Field Method to address agents' autonomous behaviour [87], ANJI [101] which is a Java implementation of the NEAT (Neuro-Evolution of Augmenting Topologies) method, Agent Analyst [38], JuPedSim [138], EPES [64], EXODUS [139], FREEgress [58], FDS+Evac [46,57], Pedsim [45].

On the other hand, some simulation models are created using native languages such as C/C++ [59,140], C# [100], Java [106], Python [48], Visual Basic [141] and i* modelling language [33], or using software packages, such as MATLAB [12] and Microsoft XNA [95]. Contrarily, 23.8% of the papers surveyed did not provide any details on what simulation software or programming language was used to construct and execute the simulation.

In summary, a wide range of simulation software and programming languages have been used to model crowd evacuation scenarios using ABM. Commercial platforms like NetLogo, AnyLogic, and Pathfinder are popular choices due to their user-friendly interfaces, built-in libraries, and visualisation capabilities. Unity and GAMA provide powerful 3D simulation and geospatial modelling features. Some researchers have also developed custom simulations using general-purpose programming languages such as C++, Java, and Python. However, the choice of technology can significantly impact simulation results due to differences in underlying algorithms and approaches. Fragmentation in modelling efforts has hindered progress, highlighting the need for standardised, modular platforms that integrate validated models, physics engines, spatial analytics, and visualisation. Developing such versatile and transferable toolkits could accelerate future research and applications in the field of crowd evacuation simulation.

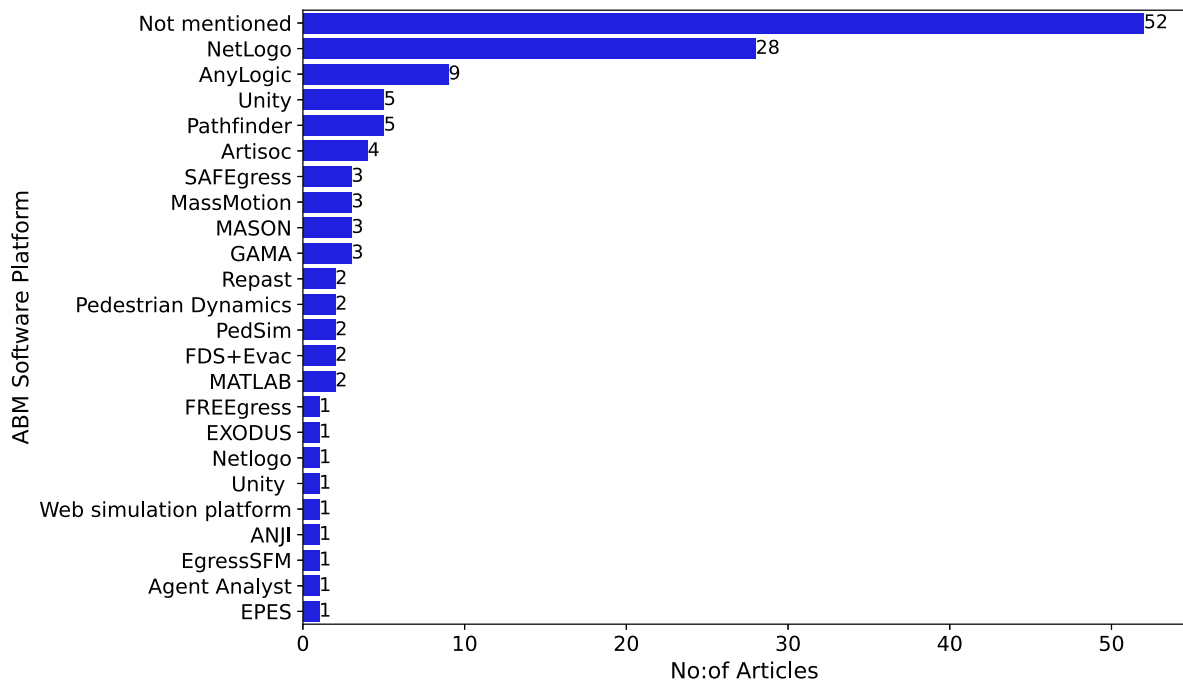


Fig. 10. ABM platforms.

3.5. Validation techniques and metrics

The main aim of crowd simulation is to create a very realistic representation of how people truly behave and interact in particular situations. Model validation is the process of evaluating how closely the model's data, calculations, and overall design accurately reflect what happens in the real world. [142] stated that models can be conceptually validated, operationally validated, validated both conceptually and operationally, or not validated at all. While conceptual validation validates underlying theories, the model development process, and the assumptions used, the operational model validation compares simulation results against a real-world scenario. The techniques used for the validation can be divided into statistical (hypothesis tests) and non-statistical (expert opinion) techniques.

The papers reviewed in this study primarily employ three distinct approaches for validating their proposed models, (1) validation against real-world experiments, (2) validation against past published data and models, (3) qualitative validation including expert opinion and Survey data. Validating models against real-world experiments is regarded as the most robust but most challenging validation technique. In this approach, the outcomes generated by the simulation are directly compared to data collected from the actual building being simulated [15,16,143], or a similar smaller space [143]. Since real evacuation data may not be available, models can be validated against drills [15,34,62,100] and Virtual Reality (VR) experiments [88] in non-emergency situations.

The most common validation technique used in the surveyed papers is published data from government sources and previous studies [17,67,87,97,98,105,119,136]. These government sources may include actual statistical data, population demographic data, witness statements, and police reports [51,136]. The validation of results is done by running statistical tests such as ANOVA [17,51]. Another common quantitative validation is sensitivity analysis to determine the effect of varying input parameters on the model output [54,103,107].

Comparing results with previously validated models is a widely accepted and highly regarded method for validating new models [95,123,128]. Sometimes researchers will create a simplified model of the simulation and validate it by comparing the results to mathematical calculations such as t-tests and queuing theory [54,144]. The study conducted by [58], compared the results from the developed model, FREEgress, with those from SAFEgress and FDS+Evac, both of which are existing and verified tools for fire evacuation simulation. Apart from the comparison with simulation tools, [145] extended the validation by following the Level 1 validation criteria from Axtell and Epstein [146], to validate against established theories. The author conducted the behaviour reproduction tests to confirm agreement with social proof theory.

Qualitative validation usually relies on the discussion of "plausible" simulation results and qualitative analysis against expert opinion [54,81,132,147,148]. Other qualitative methods compare the simulation results with well-known crowd evacuation phenomena such as the Faster-is-slower effect [99] and patterns observed in previous studies [37,83,115]. In some cases, authors qualitatively compare the data collected via post-disaster survey with the model output [13]. Additionally, [149] incorporated quantitative metrics on survey data to measure evacuation percentages and timing and regression analysis.

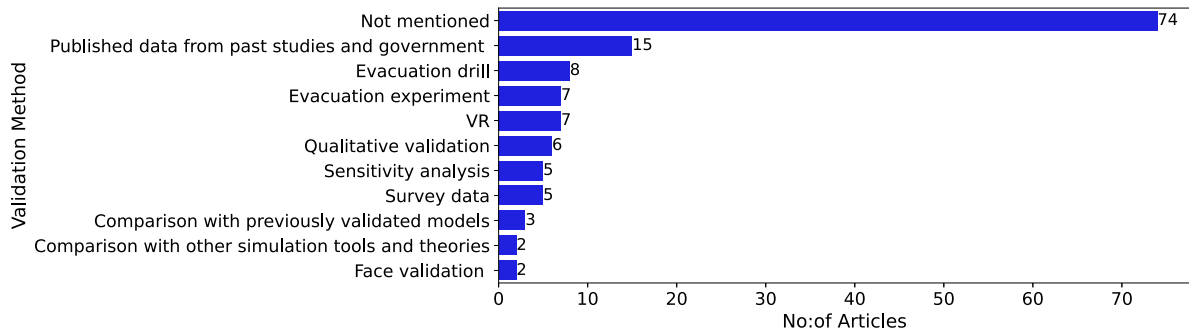


Fig. 11. Validation approaches of evacuation models.

As shown in Fig. 11, 55% of the papers failed to provide any evidence of model validation. Furthermore, nearly all of the papers omitted any reference to accessing or reproducing the model, possibly due to security concerns or the desire to maintain a competitive edge. This absence of validation and replicability raises questions about the reliability and transparency of the research presented in these papers.

Most of the papers do not provide an extensive review of validation indicators for selecting validation metrics. However, the chosen metrics align with common validation approaches seen in evacuation modelling research and the purpose and scope of the model [15,58]. Among the metrics used, the most commonly used measures is evacuation time [15,17,54,137,150], which represents the duration from the activation of an alarm until the last occupant has exited. Other metrics like travel distance, exit route choices of individuals, crowd densities, and speed have also been employed. Additionally, spatial distribution of evacuees, number of evacuated people, flow rate at exit were considered in some studies as validation metrics [15–17,54,97,137,143,145]. [136] validated their model using two metrics: consistency with observed behaviour profiles, where simulated agent behaviour was compared to six real-world resident behaviour profiles, and consistency with statistics of fatality circumstances, where simulated percentages of agents dying under different circumstances were compared to actual fatality statistics from the 2009 Victorian bushfires. The introduction of the Required Safe Egress Time (RSET) addresses a key shortcoming in identified evacuation metrics. RSET provides a more comprehensive framework by considering the full timeline of an emergency evacuation, starting from the initial detection of the threat and the sounding of the alarm [62]. It is crucial to consider at the initial stage of the evacuation.

In summary, validation is a critical aspect of developing reliable and accurate evacuation models. The reviewed studies employed various validation techniques, ranging from strong approaches, such as comparison to real-world experiments and drills, to weaker methods such as qualitative expert opinion. Many studies also validated their models against published data, previously validated models, or mathematical calculations. However, more than half of the reviewed papers did not provide evidence of model validation, raising concerns about reliability and transparency. The most common validation metrics include evacuation time, travel distance, exit route choices, crowd densities, and flow rates. While these metrics provide valuable insights, a combination of quantitative and qualitative validation methods applied across diverse emergency contexts is needed to ensure the generalisability and fidelity of evacuation models to real human behaviour.

3.6. Challenges in evacuation behaviour modelling

The development of ABM techniques in the field of evacuation decision-making has been slow and incremental. This gradual progress can be attributed to the complex challenges involved in accurately representing and simulating human behaviour with limited data across a wide range of scenarios and situations [151]. In the context of pedestrian evacuation behaviour, modern viewpoints have shifted to complex, dynamic systems composed of diverse individuals who interact with each other on multiple levels. These interactions among evacuees often involve intricate, non-linear relationships that evolve over both space and time during the evacuation process. [152]. The focus lies in modelling, encapsulating, and replicating emergent and evolving interactions within the complex system of evacuation modelling, which presents numerous challenges. Table 3 provides a summary of existing literature on the discussion of these challenges.

3.6.1. Modeller bias and limited behaviour complexity

ABMs rely heavily on the modeller's assumptions and abstractions in defining agent behaviour [14,16,17,43,47,48,50,54,71,72,88,93,97,118]. Modellers may inadvertently introduce biases based on their subjective judgements of evacuation decision-making, causing the model to inaccurately replicate complex behavioural phenomena under irrational behaviour with limited evacuation data [17,85,118]. Evacuation involves many factors interacting between individuals that cause emergent group behaviours. Hence, simple rules, such as the shortest path may not reflect human actions during evacuation with a panic mind state. ABMs are able to capture complex real human behaviours, such as changing decision situations that can incorporate adaptive behaviours [43]. Additionally, representing subjective beliefs, preferences, goals of individuals and also theory of mind for social reasoning, situational awareness, and risk perception is still lacking in almost all the reviewed articles [16,43,72]. [157] proposed that emotion, bounded

Table 3
Challenges of ABM methodology for pedestrian evacuation simulation.

Challenges	Count	Articles
Modeller bias with limited behaviour complexity	31	[43], [72], [118], [71], [14], [88], [50], [16], [17], [54], [93], [48], [97], [76], [77], [131], [85], [109], [51], [150], [60], [83], [44], [153], [117], [138], [42], [154], [104], [155], [128]
Computational Complexity and scalability	30	[43], [118], [71], [14], [88], [50], [62], [16], [17], [15], [54], [93], [48], [97], [77], [131], [85], [109], [150], [55], [83], [81], [44], [138], [156], [78], [154], [104], [155], [128]
Data Scarcity for Calibration and Validation	29	[43], [72], [118], [71], [14], [88], [50], [62], [16], [15], [54], [48], [76], [77], [131], [109], [51], [150], [60], [55], [83], [134], [44], [80], [92], [154], [104], [155], [128]
Static limited rules	12	[43], [72], [71], [50], [16], [17], [15], [150], [81], [134], [117], [92]

deliberation, and social influence (Agent Zero framework) are essential aspects in modelling cognitively plausible agents. Although the researchers try to capture more complex cognitive behaviours via hybrid approaches such as combining the OCC model and the OCEAN model [158], according to [55], these behavioural theories are not strong enough to accurately model decision making during an emergency. Carefully corroborating modelling choices with empirical data is needed to minimise speculation.

3.6.2. Computational complexity and scalability

Another common challenge faced by ABM is computational complexity [14,16,43,50,62,71,88,118]. During a hazardous event, ABMs need to handle the detailed individual behaviours of a larger crowd that interacts with the environment. This includes uncertainty in decision making, social interactions, influence of memory, learning, and emotions [159]. Computational complexity can limit model scaling [14,16,71,81,93]. Researchers are exploring methods to improve computational efficiency using hierarchical structures to model agents at different levels of abstraction, implementing agent aggregation techniques [160], and leveraging high performance computing infrastructure like multi-core processors and parallelisation [161]. Finding the right trade-off between model complexity and computational efficiency is a key challenge [131,134,136]. Achieving right scalability requires interdisciplinary collaboration to develop innovations in modelling, simulation software, and computing methods. If these challenges are overcome, rich ABMs could be leveraged for studying mass evacuation phenomena in complex real-world scenarios.

3.6.3. Lack of empirical data for model validation and calibration

Thorough calibration and validation of ABMs are essential for a meaningful output. ABMs often involve numerous heterogeneous agents with individual rule sets, necessitating extensive individual-level data [159,162]. As ABM systems grow in scale and complexity, the task of deriving significant insights from the substantial volume of data generated by these simulations becomes increasingly challenging and complex [162]. This makes calibration and validation even harder with limited real-world evacuation data [23,80,142]. Given the chaos and dangers inherent in collecting real-world behavioural data during emergency events [163], the lack of data from evacuation experiments with human intervention, including those using VR and drills, poses a significant challenge for validation due to the difficulty, expense, ethical concerns and lack of emotions involved in monitoring human behaviour under realistic emergency conditions [23]. VR experiments track lower-level behaviours show promise for calibrating and validating fine-grained agent parameters, however, these data may not replicate urgent situational factors. Expert assessments can check logical consistency, but quantitative approaches are needed. ML techniques can be used to generate synthetic data [164], and also to analyse the output of the simulation to extract patterns and validate for checking the robustness of results [164].

3.6.4. Static limited rules

ABMs traditionally define fixed decision rules or characteristics for each agent type, such as all agents follow the shortest path to the exit [43], but humans demonstrate highly diverse and context-sensitive behavioural responses dependent on individual differences and situations. Capturing this adaptive behaviour to changing situations remains an open modelling challenge [157, 165,166]. [71] suggested designing agent control rules to produce realistic global crowd behaviours. Attempts to sample from broader behavioural distributions or ensemble models with variable factors such as pattern-oriented modelling (POM) by [167] partly address this challenge. Despite these efforts, there is a growing need for more nuanced behavioural explanations to accurately capture and predict the same macro-phenomena. Traditional fixed rules often fall short in representing the complexity and variability of human behaviour, leading to oversimplified models [168]. Advancing cognitive architectures and social interaction models using evolutionary algorithms allows for the exploration of a vast space of potential behaviours, enhancing the model's ability to simulate realistic agent interactions and responses to dynamic environments.

In summary, ABMs of pedestrian evacuation behaviour face several significant challenges. Modeller bias and limited representation of behavioural complexity can lead to inaccurate replication of the complex phenomena involved in evacuation decision-making under irrational and panicked states. Computational complexity and scalability issues arise when trying to simulate the detailed individual behaviours and interactions of large crowds. Data scarcity makes it difficult to properly calibrate and validate these complex models against real-world evacuation scenarios. Finally, reliance on static, limited behavioural rules fails to capture the highly adaptive and context-sensitive nature of human responses in emergencies. Advancing ABM techniques for evacuation

simulation will require interdisciplinary efforts to develop more realistic cognitive and social models, improve computational efficiency, devise new data collection approaches, and shift towards more flexible and adaptive behavioural representations. Overcoming these challenges is crucial for ABMs to reach their full potential and provide valuable insights for evacuation planning and management.

3.7. Solutions to ABM challenges

3.7.1. Simulating complex agents with limited modeller bias

Based on the current findings from the SLR, ABMs for pedestrian evacuation have made significant strides in incorporating psychological, cognitive, and social factors to capture realistic human behaviour. Researchers have employed various strategies, such as adopting **cognitive architectures (e.g., BDI, OCEAN)** [115], and simulating social interactions (e.g., leader-follower dynamics [83,103], information sharing, emotional contagion) [82,104]. However, challenges persist in accurately representing the full complexity of human decision-making under uncertainty and validating these models against empirical data. To address these limitations, future research should focus on developing more comprehensive and flexible behavioural frameworks that integrate insights from psychology, social science, and neuroscience. This could involve leveraging advanced cognitive architectures like ACT-R or Soar, which provide a standardised approach for modelling perception, memory, learning, and decision-making processes [169,170]. Further, the established theories and models from psychology and social science, such as the **Theory of Planned Behaviour, Social Identity Theory, or the Elaboration Likelihood Model** would provide a solid foundation for representing the cognitive and social factors [171,172]. In evacuation simulations, these theories can model how intentions to evacuate are influenced by attitudes, social norms, and perceived control, represent the impact of group identities on behaviours, and inform the effectiveness of communication strategies on decision-making. Additionally, by incorporating social network dynamics and group behaviour, such as **herding, altruism, and collective intelligence, can enhance the realism of simulated evacuation scenarios** [173]. ML techniques, particularly decision trees, Bayesian networks, RL, and NN offer promising avenues for data-driven behaviour modelling, allowing agents to learn and adapt their actions based on interactions with the environment and other agents [16,101,174–177]. **Conducting sensitivity analyses and uncertainty quantification can help identify the most influential behavioural parameters and assess the robustness of model predictions under different assumptions** [178]. Moreover, engaging in interdisciplinary collaborations among behavioural scientists, modellers, and domain experts is crucial for developing empirically-grounded and contextually-relevant agent decision rules. By combining theory-driven and data-driven approaches, future ABMs can more accurately capture the adaptive and heterogeneous nature of human behaviour in emergency situations, ultimately enhancing their predictive power and practical utility for evacuation planning and management.

3.7.2. Balancing behavioural accuracy and computational efficiency

The trade-offs between incorporating complex and realistic behavioural models and the computational resources requires to simulate large-scale evacuation scenarios. **While more sophisticated behavioural models can enhance the accuracy of the simulation, they often come at the cost of increased computational complexity, leading to longer simulation times and limited scalability.** Balancing behavioural accuracy and computational efficiency in emergency evacuation simulations necessitates employing a combination of hierarchical modelling, parallel and distributed computing, model simplification techniques, hybrid modelling approaches, and ML methodologies. **Hierarchical modelling allows for the use of simplified rules for high-level behaviours while incorporating complex behaviours only for critical aspects** [179]. Parallel and distributed computing techniques, such as **Apache Spark (a distributed computing system for big data processing)** or **MPI (a message-passing system for parallel computing)**, enable the distribution of computational tasks across multiple cores or machines, thereby accelerating simulation runtime and handling large-scale evacuation scenarios [180]. Model simplification strategies, such as aggregating agents with similar characteristics or minimising detail in non-critical areas, can enhance computational efficiency while preserving essential behavioural dynamics [9]. Hybrid modelling, which combines ABMs with flow-based or cellular automata models, strikes a balance between individual-level behaviour representation and computational efficiency [150]. **ML algorithms, including deep learning and RL, capture complex human behaviour patterns and optimise real-time decision-making processes,** with methods like dimensionality reduction and transfer learning streamlining computations [12]. **Surrogate models, created using techniques such as Gaussian processes and NNs, offer simplified yet accurate representations of complex simulations, enabling rapid scenario evaluations with reduced computational demands** [181]. Further, **adaptive resolution modelling frameworks can dynamically adjust the level of detail in agent behaviours and environmental representations based on the specific requirements of the simulation scenario** [182]. This approach could involve using ML techniques to identify critical regions or time periods where high-fidelity modelling is essential, while employing simplified models in less sensitive areas to reduce computational overhead. Another potential avenue is the integration of GPU acceleration and high-performance computing infrastructures to harness the power of parallel processing for complex ABMs. Researchers should also investigate the use of advanced model reduction techniques, such as proper orthogonal decomposition or reduced basis methods, to create computationally efficient surrogates that capture the essential dynamics of the original model [183]. Additionally, the development of standardised libraries and frameworks for agent-based evacuation modelling could promote code reuse, interoperability, and performance optimisation across different simulation platforms. By exploring these innovative strategies and fostering collaboration among modellers, computer scientists, and domain experts, future research can push the boundaries of behavioural accuracy and computational efficiency in agent-based pedestrian evacuation models, ultimately enabling more realistic and scalable simulations to support emergency planning and response efforts.

3.7.3. Developing modular and scalable software architectures

One of the key challenges in agent-based pedestrian evacuation modelling is the development of software architectures that are modular, scalable, and efficient. Modular architectures allow for the flexible integration of different model components, such as agent behaviours, environmental representations, and visualisation tools. Scalability ensures that the model can handle large numbers of agents and complex environments without compromising computational performance. Component-based design allows for the decomposition of ABM software into loosely coupled, reusable components that encapsulate specific functionalities, such as agent decision-making, movement, and interaction. This modular structure facilitates the development, testing, and maintenance of individual components, as well as their integration into different evacuation scenarios [184]. Standardised interfaces and protocols, such as the Open ABM standard to promote interoperability and re-usability of model components across different platforms and implementations [185]. Efficient data structures, such as kd-trees (for organising points in multi-dimensional space) or quad-trees (for partitioning 2D space), and algorithms that minimise unnecessary computations and memory usage, such as spatial partitioning (dividing space into regions) and lazy evaluation (deferring computations until necessary), optimise the representation and processing of agent and environment data [186]. Additionally, the development of domain-specific modelling languages (DSMLs) for evacuation simulations is a promising approach. DSMLs provide high-level abstractions and intuitive syntax for describing agent behaviours, environmental dynamics, and simulation scenarios, enabling domain experts to focus on modelling rather than low-level implementation details. These languages can be supported by code generation tools that automatically translate the high-level specifications into efficient executable code, promoting modularity and reusability [187]. Another important direction is the integration of cloud computing platforms and server-less architectures to dynamically scale computational resources based on the demands of the simulation. This approach allows for the elastic provisioning of processing power and storage, ensuring that the model can handle large-scale scenarios without the need for expensive hardware investments [188]. Additionally, the development of web-based interfaces and interactive dashboards for ABM configuration, execution, and analysis can greatly enhance the accessibility and usability of evacuation simulation tools for a wider range of stakeholders, including emergency planners, decision-makers, and the general public [189]. By pursuing these innovative strategies and fostering synergies between the modelling community and software engineering experts, future research can create powerful, flexible, and user-friendly software architectures that accelerate the development and application of agent-based pedestrian evacuation models for real-world emergency preparedness and response.

3.7.4. Overcoming data limitations for validation and calibration

Addressing data scarcity in agent-based evacuation models is critical for accurate validation and calibration. Several advanced techniques can be employed to mitigate this challenge. ML techniques such as **Generative Adversarial Networks (GANs)** or **Variational Autoencoders (VAEs)** can generate synthetic datasets by learning patterns from a small set of real data [190]. Transfer learning can be utilised to adapt models trained on data from similar scenarios or locations, thereby reducing the need for extensive data collection [191]. Additionally, crowd-sourcing platforms can be leveraged to gather diverse datasets from a large number of participants in simulated evacuation scenarios, providing valuable data for model training and validation [192].

3.7.5. Evolving agent behavioural rules

Beyond numerical calibration, GAs have also been applied to evolve agent behavioural rules to discover optimal or near-optimal behaviour rules that maximise the agents' goal functions [101]. The structure optimisation using GA that can generate several ABMs for the same targeted phenomena is an emerging field in ABMs. This advancing field in GA is Inverse generative social science (IGSS). IGSS can automate the discovery of plausible explainable evacuation behavioural rules for the targeted phenomena providing accurate dataset [157,165,166]. To incorporate GAs into an ABM, the modeller needs to define the encoding scheme for the agent behaviour rules, model parameters, or model structure, as well as the fitness function that evaluates the performance of the candidate solutions. The GA then iteratively evolves the population of candidate solutions through the processes of selection, crossover, and mutation, guided by the fitness function. The best solutions found by the GA can be integrated back into the ABM to improve its accuracy, efficiency, or explanatory power.

The field of ABM for pedestrian evacuation simulations has progressed significantly, but **challenges remain in capturing human behaviour accurately, reducing computational complexity, overcoming data limitations, and developing scalable software architectures.** Integrating theories from psychology and social science, employing cognitive architectures, and leveraging ML methodologies can enhance the realism and accuracy of agent decision-making processes. Techniques such as surrogate modelling, hierarchical modelling, parallel computing, model simplification, hybrid approaches, and dimensionality reduction offer solutions for balancing behavioural fidelity with computational efficiency, while generative models, transfer learning, and GAs address data scarcity and optimal behaviour rule discovery.

4. Future directions

The findings of this SLR reveal significant progress in the development and application of ABMs for pedestrian evacuation simulation. However, several challenges persist, such as the **need for more realistic behavioural representations, robust validation techniques, and scalable computational frameworks.** To address these limitations and advance the field, future research should focus on:

– Advanced Cognitive and Social Models:

- Integrating insights from behavioural science and psychology to develop more comprehensive and empirically-grounded agent decision-making models. Developing comprehensive frameworks can be achieved by combining available theories to enhance the accuracy of behaviour representation.
- Systematic comparisons of different theoretical frameworks to identify their strengths, limitations, and complementarity in the context of emergency evacuations.
- Explore how to dynamically adapt behavioural theories based on situational factors and individual differences among agents.
- Incorporating dynamic social network analysis in evacuation models can reveal insights into group behaviours and information dissemination during emergencies. Future research can focus on real-time adaptation of evacuation models based on live social media and communication data.

– Machine Learning and Data-Driven Approaches:

- Leveraging ML techniques, such as RL and NN, to enable adaptive and data-driven agent behaviours. Research can focus on creating adaptive learning environments where agents continually update their decision-making processes based on new data.
- Developing techniques for extracting interpretable rules from trained models and validating their transferability to unseen contexts is crucial for building trust in ML-driven ABMs among domain experts and end-users.
- Validating synthetic datasets generated utilising GANs and VAEs through comparison with real evacuation drills.
- Developing robust transfer learning frameworks that can seamlessly adapt models trained on different but related evacuation scenarios to new contexts. Research can focus on the generalisation capabilities of these models and their performance in diverse evacuation environments.

– Computational Efficiency and Scalability:

- Creating multi-level hierarchical models that can switch between detailed and simplified agent behaviours based on the simulation context. Research can focus on optimising the decision-making process for these switches to balance computational load and accuracy.
- Leveraging advancements in cloud computing and edge computing to distribute computational tasks efficiently. Future research can explore the integration of real-time data processing capabilities to enhance the responsiveness of evacuation simulations.
- Developing advanced surrogate models using techniques like Bayesian optimisation to reduce computational demands while maintaining model fidelity. Research can focus on the applicability of these models across different scales of evacuation scenarios.

– Interdisciplinary Collaboration and Empirical Validation:

- Establishing interdisciplinary research teams to co-develop ABMs that are grounded in empirical data and behavioural science theories and creating standardised validation protocols and benchmark datasets to facilitate model comparison, calibration, and verification.
- Explore innovative data collection methods, such as participatory sensing, social media mining, and virtual reality experiments, to gather high-resolution behavioural data during actual or simulated emergencies. Establishing data sharing platforms and standardised datasets within the community can also accelerate progress.
- Enhancing the realism of VR-based evacuation drills by incorporating physiological and emotional responses of participants. Research can explore the use of biometric data to validate and calibrate agent behaviours in simulations.
- Developing robust methods for sensitivity analysis and uncertainty quantification that can handle high-dimensional parameter spaces. Future studies can aim at automating these processes to provide real-time feedback during simulations.

– Modular and Scalable Software Architectures:

- Establish community-wide standards, benchmark models, and open-source libraries for modular, scalable software architectures that facilitate the sharing, reuse, and integration of model components across different platforms and application domains.
- Exploring the integration of server-less computing paradigms to dynamically scale simulation workloads. Future research can investigate the cost-efficiency and performance benefits of these architectures in large-scale evacuation scenarios.
- Advancing DSMLs to support more complex and customisable agent behaviours and creating intuitive and user-friendly interfaces that allow domain experts to easily modify and extend model components
- Developing interactive dashboards and web-based interfaces that provide real-time insights into simulation progress and outcomes. Research can focus on user experience design to enhance accessibility for non-technical stakeholders.

– Exploring Evolutionary Algorithms:

- Integrating domain knowledge into the evolutionary process and developing metrics for assessing the behavioural validity of evolved rules using IGSS are important directions for future research.

- Future research can focus on integrating GAs with real-time data to continuously improve agent behaviours during ongoing simulations and explore the scalability of IGSS for large-scale evacuation scenarios and its applicability to different emergency contexts.

By addressing these challenges and opportunities, the field of agent-based pedestrian evacuation modelling can continue to advance towards more realistic, efficient, and actionable simulations that inform emergency preparedness, response, and resilience efforts worldwide.

5. Conclusion

This systematic review provides the most comprehensive synthesis to date regarding agent-based modelling approaches for simulating pedestrian behaviour during emergency evacuations. By rigorously analysing 134 studies spanning models, methods, tools, validation techniques, and challenges from 2013 to 2023, the review offers an up-to-date and critical assessment of the state-of-the-art in this field. Unlike previous reviews that have broadly surveyed the ABM evacuation modelling landscape, this work delves deeply into the methodological foundations, decision-making frameworks, and computational architectures used in existing models.

The review uncovers several key findings and patterns in the application of ABMs for pedestrian evacuation. Influential papers published in high-impact journals such as *Safety Science* and the *International Journal of Disaster Risk Reduction* have made significant strides in integrating psychological and cognitive theories, realistic environmental representations, and hybrid modelling approaches. In recent years, there has been a growing focus on simulating evacuations in complex urban environments and large-scale emergencies beyond just building-level scenarios. Modelling of agent characteristics has increasingly incorporated demographic, psychological, and social factors, although capturing the full complexity of human behaviour remains challenging. While diverse agent architectures, decision-making models, and movement algorithms have been employed, there is a predominance of simplistic rule-based approaches that lack flexibility and empirical grounding. Simulation technology has evolved with the use of advanced platforms like AnyLogic and GAMA, but issues persist around computational efficiency and model composability. Validation techniques have primarily relied on limited real-world experiments, drills, and expert judgement, with a notable scarcity of quantitative benchmarks and standardised protocols. The review identifies modeller bias, computational complexity, validation data, and static limited rules as major obstacles confronting current ABMs while also highlighting promising avenues such as ML, surrogate modelling, and evolutionary computation to enhance behavioural fidelity and scalability.

The application of advancements in ABM approaches in evacuation modelling is essential with the development of more sophisticated behavioural frameworks, data-driven rule generation techniques, and high-performance computing architectures, which hold significant promise for enhancing the realism and reliability of evacuation simulation tools. By integrating data-driven rule generation techniques such as IGSS with more accurate evacuation data, the next generation of ABMs could provide emergency managers with unprecedented decision support capabilities [165] and accurately capture the diverse and adaptive decision-making processes of human agents under duress. Moreover, the utilisation of high-performance computing platforms allows for the analysis of evacuation dynamics at realistic scales [164]. These advancements can ultimately enable the development of powerful simulation tools that allow emergency planners to virtually assess and optimise evacuation strategies, identify potential bottlenecks, and stress-test response plans against a wide range of disaster scenarios. Such sophisticated ABMs could serve as digital twins of real-world environments, providing interactive virtual laboratories for training responders, evaluating building designs, and even personalising evacuation instructions in real-time based on situational conditions. As these modelling techniques continue to mature and become more accessible through interdisciplinary collaboration and open platforms, they hold immense potential to be translated into practical decision support systems that can meaningfully enhance the effectiveness of emergency response and save lives during crisis situations.

However, it is important to acknowledge the limitations of this review. The analysis is based on a specific set of 134 papers published in the Scopus database between 2013 and 2023. While this approach ensures a comprehensive coverage of the recent literature, it may not exhaustively capture all relevant studies, especially those published in other databases, in grey literature, or outside the specified time-frame. Additionally, the review focuses primarily on ABM advancement in pedestrian evacuation modelling, and the findings may not provide a comprehensive overview of the current state of the art in emergency pedestrian evacuation research as a whole or directly generalisable to other types of emergency scenarios, such as vehicle-based evacuations or multi-modal transportation contexts. Future reviews could expand the search criteria, consider a longer time horizon, and include a broader range of emergency situations to provide a more comprehensive understanding of ABM applications in disaster management.

In conclusion, this systematic review underscores the immense potential of ABM as a powerful tool for simulating and understanding pedestrian behaviour during emergency evacuations. By harnessing the strengths of ABMs in capturing complex individual and collective dynamics, researchers can generate valuable insights into the factors that influence evacuation processes and outcomes. **To guide future research and development efforts, researchers should prioritise the collection and sharing of high-quality empirical data, adopt modular and open-source development practices, invest in advanced computational techniques, and engage with stakeholders to ensure alignment with real-world needs and evidence-based decision-making.** As the frequency and intensity of natural and human-induced disasters continue to rise in the face of climate change and urbanisation, investing in the advancement of agent-based pedestrian evacuation models becomes increasingly critical. With the right tools, knowledge, and collaboration, we can create more resilient and adaptive built environments that prioritise the safety and well-being of occupants during emergencies. This review serves as a clarion call for researchers, practitioners, and policymakers to unite in this endeavour and work towards a future where no life is lost due to inadequate evacuation planning and response.

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Declaration of competing interest

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Data availability

Data will be made available on request.

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